Year 13 Further Maths - Geometric distribution

Similar questions - Modelling, variance and goodness of fit

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The random variable D has the distribution Geo(p) with Var$(D) = 6$.

(a) Determine Var$(2D+3)$. **[1]**

(b) Determine p and hence E$(2D+3)$. **[6]**

(c) Find P$(D > \text{E}(D))$. **[3]**

2 In a large collection of coloured beads of identical size the proportion of gold beads is p. One bead is chosen at random, its colour noted and it is replaced. This is repeated until a gold bead is chosen. The first gold bead is the Xth bead chosen.

(a) You are given that $p = 0.25$.

(i) Find P$(3 \leqslant X \leqslant 8)$. **[2]**

(ii) Determine the smallest n for which P$(X = n) < 0.05$. **[2]**

(b) You are given instead that Var$(X) = 30$. Determine E(X). **[5]**

3 A researcher interviews people at random, one after another, until finding one who supports a particular policy; the number interviewed that day is R. Assume that 2% of the population support the policy.

(a) State an appropriate distribution for R, giving any parameters, and find P$(30 < R \leqslant 90)$. **[3]**

(b) The researcher believes that the variance of any discrete random variable X is $[\text{E}(X)]^2 - \text{E}(X)$. Show that this is correct for the distribution in part (a). **[3]**

(c) Show that the researcher's formula is not correct for a general random variable by considering a variable which takes the value 2 with probability 1. **[2]**

4 A librarian looks through books in turn to find one referring to a particular topic. She believes that $\frac{1}{12}$ of the books refer to the topic. The first reference is found in the Xth book examined.

(a) A student says that X cannot be geometric because the number of books is finite. Explain whether you agree. **[1]**

(b) State one modelling assumption needed for a geometric model, and suggest a reason why it may not hold. **[2]**

(c) Find the smallest n such that the probability she needs to look at no more than n books exceeds 0.95. **[3]**

(d) She needs to find four different books referring to the topic. Find the probability that the fourth is the 40th book she examines. **[2]**

5 A spinner has the number 1 on one quarter of its surface. Huda spins it repeatedly and counts the number of spins X up to and including the first 1. She does this 80 times; the results are shown below.

x	1	2	3	4	5	$\geqslant 6$
Frequency	22	15	12	9	6	16

(a) State a suitable distribution for X and write down its parameter. **[2]**

(b) Show that the expected frequency for $x \geqslant 6$ is 18.99 (to 2 d.p.) **[2]**

(c) Carry out a goodness of fit test at the 5% level. **[6]**

Question 1

(a) $\mathrm{Var}(2D+3) = 4 \times 6 = 24$

(b) $\dfrac{1-p}{p^2} = 6 \Rightarrow 6p^2 + p - 1 = 0 \Rightarrow p = \frac{1}{3}$ ($p > 0$). $\mathrm{E}(D) = 3$, $\mathrm{E}(2D+3) = 9$

(c) $\mathrm{P}(D > 3) = \left(\frac{2}{3}\right)^3 = \frac{8}{27} = 0.296$

Question 2

(a)(i) $\mathrm{P}(X \geqslant 3) - \mathrm{P}(X \geqslant 9) = 0.75^2 - 0.75^8 = 0.462$

(ii) $0.25(0.75)^{n-1} < 0.05 \Rightarrow (n-1)\ln 0.75 < \ln 0.2$, $n - 1 > 5.59$, $n = 7$

(b) $\dfrac{1-p}{p^2} = 30 \Rightarrow 30p^2 + p - 1 = 0$, $p = \dfrac{-1+11}{60} = \frac{1}{6}$; $\mathrm{E}(X) = 6$

Question 3

(a) $R \sim \mathrm{Geo}(0.02)$. $\mathrm{P}(30 < R \leqslant 90) = 0.98^{30} - 0.98^{90} = 0.383$

(b) $\mathrm{E} = \frac{1}{p}$, $\mathrm{Var} = \frac{1-p}{p^2} = \frac{1}{p^2} - \frac{1}{p} = \mathrm{E}^2 - \mathrm{E}$ (as required)

(c) A constant has $\mathrm{Var} = 0$ but $\mathrm{E}^2 - \mathrm{E} = 4 - 2 = 2$

Question 4

(a) Disagree: a geometric model is an approximation; with many books and small p the chance of needing more than the number available is negligible (or the model is for a very large collection)

(b) Each book independently has the same probability $\frac{1}{12}$; not valid if topics cluster on shelves

(c) $1 - \left(\frac{11}{12}\right)^n > 0.95 \Rightarrow n > 34.4$, so $n = 35$

(d) $\binom{39}{3}\left(\frac{1}{12}\right)^4\left(\frac{11}{12}\right)^{36} = 0.0192$

Question 5

(a) $X \sim \mathrm{Geo}(0.25)$

(b) $\mathrm{P}(X \geqslant 6) = 0.75^5 = 0.2373$; $80 \times 0.2373 = 18.99$

(c) H_0: $\mathrm{Geo}(0.25)$ is a suitable model; H_1: it is not. Expected: 20, 15, 11.25, 8.44, 6.33, 18.99; contributions 0.2, 0, 0.05, 0.04, 0.02, 0.47 giving $\chi^2 = 0.77$. Degrees of freedom $6 - 1 = 5$; critical value 11.07. $0.77 < 11.07$ so do not reject H_0: the geometric model is consistent with the data